

Implementation report: α -attractor inflaton models in COSMOLATTICE

Covers arXiv:2510.18656 (Ellis, Garcia, Olive, Verner)

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Abstract

This report documents the COSMOLATTICE implementation of every inflationary model found viable in arXiv:2510.18656: the generalized and deformed α -attractor E-models and T-models. It lists the new files, explains how the model headers and the initial-condition helper were built, and justifies every parameter choice in the benchmark parameter files. All families are covered by two model headers, `attractorE.h` and `attractorT.h`, plus 14 parameter files. It also derives the expressions used to redshift the lattice GW spectra and wavenumbers to today's frequency and $h^2\Omega_{\text{GW}}$ (Sec. 8). The physics background (data constraints, derivative formulas, analytic approximations) is in the companion note `notes/inflation_models.tex`.

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1 Summary

- **Two model headers** cover the four viable families. A deformed model with $\kappa = 1$ is exactly the corresponding generalized model, and $k = 2$, $\alpha = 1$, $\kappa = 1$ of the E-model is Starobinsky. α , k , κ , λ and the daughter coupling g are read at run time, so no recompilation is needed to scan them.
- **One Python helper** (`notes/attractor_ics.py`) solves the homogeneous background. Given (α, k, κ, N_*) it returns the CMB normalization λ and the inflaton amplitude and velocity at the end of inflation.
- **14 benchmark parameter files** reproduce the points quoted in the paper.
- Both models compile against COSMOLATTICE 2.0 and ran without errors in short tests (Sec. 7).

2 New and modified files

2.1 New files

File	Content
<code>models/attractorE.h</code>	Generalized + deformed E-model (α -Starobinsky), inflaton + daughter field.
<code>models/attractorT.h</code>	Generalized + deformed T-model, inflaton + daughter field.
<code>models/parameter-files/attractorE_*.in</code> (8 files)	E-model benchmarks (Table 1).
<code>models/parameter-files/attractorT_*.in</code> (6 files)	T-model benchmarks (Table 1).
<code>notes/attractor_ics.py</code>	Background solver: λ , φ_i , $\dot{\varphi}_i$ from (α, k, κ, N_*) .
<code>notes/inflation_models.tex</code> (+ PDF)	Physics note: which models, why, potentials, derivatives, benchmark table.
<code>notes/implementation_report.tex</code> (+ PDF)	This report.
<code>runs/E_starobinsky_gw/</code> , <code>runs/T_k2_a1_gw/</code>	Preheating runs with GWs (<code>run.in</code> + output).
<code>runs/attractor_runs.ipynb</code> , <code>runs/make_notebook.py</code>	Analysis notebook and its generator; Sec. 5 redshifts the GW spectra to today (Sec. 8).

Two build directories, `build_attractorE/` and `build_attractorT/`, were also created. They are ignored by git.

2.2 Modified files

No tracked source file of COSMOLATTICE was modified: the new models are self-contained headers picked up by the existing CMake option `-DMODEL=<name>`.

Before this work the repository was updated to the latest upstream version (COSMOLATTICE 2.0, commit `acc8278d`). Two local changes to tracked files (deletion of `src/models/parameter-files/lphi4.in`, edit of `lphi4U1.in`) were reverted to the upstream versions. They are saved in git `stash` entry “local changes before pulling v2.0”. Your untracked files (`src/models/lphi4_only.h`, `m2phi2.h`, their `.in` files, `papers/`) were not touched. In v2.0 models live in `models/`, not `src/models/`.

3 Models implemented

With $x \equiv \varphi/M_P$, $b \equiv \sqrt{2/(3\alpha)}$, $c \equiv b/2 = 1/\sqrt{6\alpha}$ and $V_0 = \frac{3}{4}\lambda M_P^4$:

$$\text{attractorE: } V = V_0 [\kappa(1 - \cosh bx) + \sinh bx]^k, \quad (1)$$

$$\text{attractorT: } V = V_0 \left[\frac{1}{2}(1 + \kappa + (1 - \kappa) \cosh bx) \right]^2 \tanh^k(cx). \quad (2)$$

These are the deformed models of the paper (its Sec. V.B, “Deformed No-Scale Attractors”). They reduce to

- $\kappa = 1$: generalized E-model $V_0(1 - e^{-bx})^k$ and generalized T-model $V_0 \tanh^k(cx)$ (paper Sec. V.A);
- $\kappa = 1$, $k = 2$: standard E- and T-models; additionally $\alpha = 1$: Starobinsky.

We checked analytically that the paper’s superpotential construction (its Appendix A) gives the bracket in Eq. (1) raised to the power k . Near the minimum both potentials behave as φ^k for any κ .

A second scalar χ (field index 1) is coupled through $\frac{1}{2}g^2\varphi^2\chi^2$, as in the existing `lphi4.h` and `tanh2.h`. The default is $g = 0$: then χ is a free massless spectator and the run studies the inflaton’s self-resonance only.

4 How the model headers were built

The headers follow the structure of the existing `models/tanh2.h` and `models/lphi4.h`: a `ModelPars` struct (2 scalars, 2 potential terms, `double`), a constructor that reads parameters and sets the program-variable rescaling, and the functions `potentialTerms`, `potDeriv`, `potDeriv2` for each field.

4.1 Run-time parameters

Key	Default	Symbol	Meaning
<code>lambda</code>	required	λ	CMB normalization, $V_0 = \frac{3}{4}\lambda M_P^4$
<code>alphaAtt</code>	1	α	attractor parameter. Not called <code>alpha</code> , which COSMOLATTICE already uses for the time-rescaling exponent
<code>k</code>	2	k	power at the minimum; must be an even integer ≥ 2 (checked, else the run stops with <code>RunParametersInconsistent</code>)
<code>kappa</code>	1	κ	plateau deformation
<code>g</code>	0	g	inflaton–daughter coupling
<code>initial_amplitudes</code>	required	φ_i, χ_i	GeV
<code>initial_momenta</code>	0 0	$\dot{\varphi}_i, \dot{\chi}_i$	GeV ²

Since k is a run-time number, powers are written as `pow(expr, k)` on field expressions. TempLat supports this, and an integer-valued floating exponent is well defined for negative bases, which occur for $x < 0$.

4.2 Program variables

COSMOLATTICE evolves $\tilde{\varphi} = \varphi/f_*$ in the time variable $d\tilde{\eta} = a^{-\alpha_t}\omega_* dt$. The program potential is $\tilde{V} = V/(f_*^2\omega_*^2)$. We chose:

$$f_* = M_P, \quad \omega_* = \sqrt{\lambda_k} M_P x_i^{(k-2)/2}, \quad \alpha_t = \frac{3(k-2)}{k+2}, \quad (3)$$

where $x_i = |\varphi_i|/M_P$ and λ_k is defined by the small-field limit $V \simeq (\lambda_k/k)M_P^{4-k}\varphi^k$:

$$\lambda_k^{(E)} = k \frac{3}{4} \lambda b^k, \quad \lambda_k^{(T)} = k \frac{3}{4} \lambda c^k. \quad (4)$$

Why these choices.

- $f_* = M_P$: the program field is $x = \varphi/M_P$, so the potential is written directly in terms of the functions of Sec. 3, with b, c unchanged. The plateau and the minimum are both $\mathcal{O}(1)$ in these units.
- ω_* is the oscillation frequency of a condensate of amplitude φ_i in the φ^k potential. For $k = 2$ it is the inflaton mass: $\omega_* = m_\varphi = \sqrt{\lambda/\alpha} M_P$ (E), $\frac{1}{2}\sqrt{\lambda/\alpha} M_P$ (T). For $k = 4$ it reduces to the `lphi4.h` choice $\omega_* = \sqrt{\lambda_4} \varphi_i$. One oscillation therefore takes $\Delta\tilde{\eta} \sim 2\pi$ at the start.
- α_t : in a φ^k potential the amplitude decays as $a^{-6/(k+2)}$, so the oscillation frequency decays as $a^{-3(k-2)/(k+2)}$. With $\alpha_t = 3(k-2)/(k+2)$ the frequency stays constant in program time. This gives $\alpha_t = 0$ for $k = 2$ (as in `tanh2.h`) and $\alpha_t = 1$ for $k = 4$ (as in `lphi4.h`); $\alpha_t = 3/2$ for $k = 6$, $\alpha_t = 2$ for $k = 10$.

With these choices, the code computes two constants once in the constructor:

$$\text{pref} = \frac{3}{4} \lambda \frac{M_P^2}{\omega_*^2}, \quad \text{qInt} = \frac{g^2 M_P^2}{\omega_*^2}, \quad (5)$$

so that $\tilde{V} = \text{pref} \cdot F(x) + \frac{1}{2} \text{qInt} x^2 \tilde{\chi}^2$, with $F = V/V_0$ the bracket expressions of Eqs. (1)–(2).

4.3 Numerically safe forms

Near the minimum $\kappa(1 - \cosh bx)$ and $(1 - \kappa)(\cosh bx - 1)$ suffer from cancellation. The code uses the identity $\cosh y - 1 = 2 \sinh^2(y/2)$:

$$G \equiv \kappa(1 - \cosh bx) + \sinh bx = \sinh bx - 2\kappa \sinh^2(bx/2), \quad (6)$$

$$D \equiv \frac{1}{2}(1 + \kappa + (1 - \kappa) \cosh bx) = 1 + (1 - \kappa) \sinh^2(bx/2). \quad (7)$$

The derivatives implemented are

$$\text{E: } F' = kG^{k-1}G', \quad F'' = k[(k-1)G^{k-2}G'^2 + G^{k-1}G''], \quad (8)$$

$$G' = b(\cosh bx - \kappa \sinh bx), \quad G'' = b^2(\sinh bx - \kappa \cosh bx), \quad (9)$$

$$\text{T: } F' = 2DD'T + D^2T', \quad F'' = 2(D'^2 + DD'')T + 4DD'T' + D^2T'', \quad (10)$$

with $D' = \frac{1}{2}(1 - \kappa)b \sinh bx$, $D'' = \frac{1}{2}(1 - \kappa)b^2 \cosh bx$, $T = \tanh^k(cx)$, $T' = kc \tanh^{k-1}(cx) \text{sech}^2(cx)$, $T'' = kc^2 \tanh^{k-2}(cx) \text{sech}^2(cx) [(k-1) \text{sech}^2(cx) - 2 \tanh^2(cx)]$. All of them were checked against finite differences of the potential. The second derivatives set the initial effective masses through `setInitialPotentialAndMassesFromPotential()`.

5 The initial-condition helper

`notes/attractor_ics.py` needs `numpy` and `scipy`. These are available in the `cosmolattice` conda environment.

Usage.

```
~/miniconda3/envs/cosmolattice/bin/python3 notes/attractor_ics.py E --alpha 1 --k 2 --
kappa 1 --nstar 50
```

It prints `lambda`, `alphaAtt`, `k`, `kappa`, `initial_amplitudes` and `initial_momenta` in the `.in` syntax, plus a comment line with the slow-roll n_s , r , x_* , x_{end} and H_{end} .

Method.

1. The background equation is integrated in e -folds N , where the trajectory $x(N)$ does not depend on λ :

$$x'' = -(3 - \varepsilon_H) \left(x' + \frac{F'(x)}{F(x)} \right), \quad \varepsilon_H = \frac{1}{2} x'^2. \quad (11)$$

It starts on the slow-roll attractor, $x' = -F'/F$, at a field value x_0 increased until the slow-roll estimate gives at least $N_* + 15$ e -folds.

2. The end of inflation is the event $\varepsilon_H = 1$. This gives x_{end} and x'_{end} .
3. x_* is the value N_* e -folds before the end, read from the dense ODE solution.
4. λ is fixed from $A_s = V_*/(24\pi^2 \varepsilon_{V_*} M_P^4)$ with $A_s = 2.1 \times 10^{-9}$, i.e. $\lambda = 32\pi^2 A_s \varepsilon_{V_*}/F(x_*)$.
5. The initial conditions are set at the end of inflation:

$$\varphi_i = x_{\text{end}} M_P, \quad \dot{\varphi}_i = M_P H_{\text{end}} x'_{\text{end}}, \quad H_{\text{end}}^2 = \frac{V(x_{\text{end}})}{M_P^2(3 - \varepsilon_H)}. \quad (12)$$

The ODE uses `rtol` = 10^{-10} . The quoted n_s , r are slow-roll estimates, accurate to $\sim 10^{-3}$ in n_s ; the paper's values come from solving the Mukhanov–Sasaki equation.

Check against the paper. For Starobinsky with $N_* = 55$ the helper gives $\lambda = 1.57 \times 10^{-10}$ (paper: 1.6×10^{-10}), $x_* = 5.32$ (paper: 5.35), $x_{\text{end}} = 0.615$ (paper's analytic estimate: 0.63), $n_s = 0.964$, $r = 0.0037$ (paper, exact: 0.9649, 0.00355).

6 Choice of the parameters

6.1 Model parameters (α, k, κ)

The benchmarks are the parameter points for which the paper quotes predictions or which it identifies as best fits (its Results section, collected in Table 2 of `inflation_models.tex`):

- reference points: Starobinsky ($k = 2$, $\alpha = 1$) and the $k = 2$ T-model;
- steeper minima, which raise N_* and n_s through a stiffer reheating equation of state $w = (k - 2)/(k + 2)$: $k = 4, 6$ (E), $k = 6, 10$ (T) at $\alpha = 1$, and $\alpha = 5$ for E with $k = 6, 10$, where the paper finds overlap with the ACT region;
- deformed plateaus: $\kappa = 0.9999$ for $k = 2$, $\kappa = 0.99999$ for $k = 4, 6$ (see below), and the deformed E-model $k = 6$, $\alpha = 4$ point.

The value of κ for $k = 4, 6$. The paper's text says $\kappa = 0.9999$ for every deformed benchmark. With that value the $k = 4, 6$ points give a blue tilt ($n_s = 0.99$ – 1.04), far from the quoted $n_s \simeq 0.966$ – 0.970 . With $\kappa = 0.99999$ the helper reproduces them: E $k = 4$: $n_s = 0.969$, $r = 0.0044$ (paper 0.970, 0.004); T $k = 4$: $n_s = 0.966$, $r = 0.0042$ (paper 0.966, 0.004); T $k = 6$: $n_s = 0.968$ (paper 0.969). The corresponding figure file in the arXiv source is also named `fulltplots_kappa0p99999.pdf`, so we used $\kappa = 0.99999$ for these three files. The deformed E $k = 6$, $\alpha = 4$ point does match $\kappa = 0.9999$ ($n_s = 0.973$, $r = 0.017$; paper 0.974, 0.016), so it keeps that value. The paper's $r \simeq 0.012$ for deformed T $k = 6$ is not reproduced for either κ (we get 0.004).

6.2 Number of e -folds N_*

N_* is an input of the helper and fixes λ and the slow-roll observables. It does not change the lattice initial conditions much: x_{end} does not depend on N_* , while λ and $\dot{\varphi}_i$ change at the 10% level. Values were taken from the paper for its conservative reheating window $10^2 \text{ GeV} \lesssim T_{\text{RH}} \lesssim 10^{10} \text{ GeV}$:

- $k = 2$: $N_* = 50$, the value the paper uses for reference; it corresponds to $T_{\text{RH}} \simeq 5 \times 10^7 \text{ GeV}$ for Starobinsky. From the paper’s App. B formula $N_* \simeq 59.55 - \frac{1}{3} \ln N_* + \frac{1}{3} \ln(T_{\text{RH}}/M_P)$, $N_* \simeq 45.7\text{--}51.8$ across the window.
- $k = 4$: $N_* = 55.7$ (E), 55.8 (T). For $w = 1/3$, N_* does not depend on T_{RH} (paper, App. B).
- $k = 6$: $N_* = 57\text{--}58$, the range quoted by the paper.
- $k = 10$: $N_* = 60$, middle of the paper’s 59–61.5.

For $k \geq 6$ the true N_* depends on the inflaton fragmentation temperature, which is exactly what the lattice runs can measure. These values should be revisited once that is known.

6.3 Lattice and run settings

All 14 files share the same settings, which are untuned starting points:

Key	Value	Reason
<code>expansion</code>	<code>true</code>	self-consistent FRW expansion; needed for reheating
<code>evolver</code>	<code>VV2</code>	second-order velocity Verlet, as in <code>tanh2.in</code>
<code>N</code>	64	modest grid for exploration; 32 was used for the tests
<code>kIR</code>	0.05	$H_{\text{end}}/\omega_* \simeq 0.24$ for Starobinsky (the measured initial Friedmann value in program units, $0.058 \simeq 0.24^2$), so the box contains modes from above to well below the horizon scale
<code>dt</code>	0.02	~ 300 steps per initial oscillation period (2π); much smaller than the lattice spacing $\delta x = 2\pi/(N k_{\text{IR}}) \simeq 2$
<code>kCutOff</code>	2	initial vacuum fluctuations only below $\tilde{k} = 2$, inside the lattice range ($\tilde{k}_{\text{max}} \simeq \sqrt{3} N k_{\text{IR}}/2 \simeq 2.8$)
<code>tMax</code>	200	~ 30 oscillations; extend for fragmentation studies
<code>tOutput*</code>	0.5 / 5 / 20	frequent / infrequent / rare measurements
<code>g</code>	0	self-resonance only; set $g > 0$ to study preheating into χ
<code>withGWs</code>	<code>false</code>	off for the first runs

Before production runs, check resolution by varying `N`, `kIR` and `dt`, and watch the Friedmann-constraint output.

6.4 Benchmark files

7 Validation

1. **Derivatives:** F' and F'' of all four families were compared with finite differences at a generic point ($k = 6$, $\kappa = 0.93$); agreement to $\sim 10^{-8}$.
2. **Compilation:** both headers build against COSMOLATTICE 2.0 (TempLat v1.0.2) with the `cosmolattice` conda environment (cmake 4.2.3, FFTW3).
3. **Short runs** ($N = 32$, $\tilde{t} \leq 20$):

File (<code>attractor...in</code>)	α	k	κ	N_*	λ	φ_i [GeV]	$\dot{\varphi}_i$ [GeV ²]	(n_s, r) slow roll
<code>E_k2.a1_starobinsky</code>	1	2	1	50	1.89×10^{-10}	1.50×10^{18}	-2.79×10^{31}	0.9604, 0.0044
<code>E_k4.a1</code>	1	4	1	55.7	1.60×10^{-10}	2.68×10^{18}	-2.29×10^{31}	0.9639, 0.0038
<code>E_k6.a1</code>	1	6	1	57	1.56×10^{-10}	3.53×10^{18}	-2.14×10^{31}	0.9645, 0.0037
<code>E_k6.a5</code>	5	6	1	58	7.23×10^{-10}	5.16×10^{18}	-2.16×10^{31}	0.9653, 0.0155
<code>E_k10.a5</code>	5	10	1	60	7.09×10^{-10}	7.37×10^{18}	-1.83×10^{31}	0.9660, 0.0152
<code>E_k2.a1_kappa0p9999</code>	1	2	0.9999	50	2.61×10^{-10}	1.50×10^{18}	-3.28×10^{31}	0.9737, 0.0062
<code>E_k4.a1_kappa0p99999</code>	1	4	0.99999	55.7	1.86×10^{-10}	2.68×10^{18}	-2.46×10^{31}	0.9694, 0.0044
<code>E_k6.a4_kappa0p9999</code>	4	6	0.9999	57	7.54×10^{-10}	4.93×10^{18}	-2.51×10^{31}	0.9729, 0.0169
<code>T_k2.a1</code>	1	2	1	50	2.08×10^{-10}	2.04×10^{18}	-2.44×10^{31}	0.9591, 0.0049
<code>T_k6.a1</code>	1	6	1	58	1.55×10^{-10}	4.83×10^{18}	-1.92×10^{31}	0.9646, 0.0037
<code>T_k10.a1</code>	1	10	1	60	1.45×10^{-10}	6.30×10^{18}	-1.84×10^{31}	0.9658, 0.0034
<code>T_k2.a1_kappa0p9999</code>	1	2	0.9999	50	2.78×10^{-10}	2.04×10^{18}	-2.82×10^{31}	0.9714, 0.0066
<code>T_k4.a1_kappa0p99999</code>	1	4	0.99999	55.8	1.80×10^{-10}	3.71×10^{18}	-2.11×10^{31}	0.9659, 0.0042
<code>T_k6.a1_kappa0p99999</code>	1	6	0.99999	57	1.80×10^{-10}	4.83×10^{18}	-2.07×10^{31}	0.9681, 0.0042

Table 1: The 14 benchmark parameter files. All numbers come from `notes/attractor_ics.py`; the generating command is recorded in the header of each file.

Parameter file	NaN	Friedmann rel. error at $\tilde{t} = 19.5$	Behaviour
<code>attractorE_k2.a1_starobinsky</code>	none	2.3×10^{-4}	damped oscillation around $x = 0$
<code>attractorT_k2.a1</code>	none	3.8×10^{-4}	damped oscillation
<code>attractorE_k6.a1</code>	none	4.8×10^{-6}	anharmonic, asymmetric oscillation
<code>attractorE_k6.a4_kappa0p9999</code>	none	1.1×10^{-5}	anharmonic oscillation
<code>attractorT_k6.a1_kappa0p99999</code>	none	2.6×10^{-5}	anharmonic, symmetric oscillation

The initial program momentum of the Starobinsky run, $\tilde{\pi} = -0.342$, agrees with the hand conversion $\dot{\varphi}_i/(f_*\omega_*)$.

Not yet done: comparison of `attractorT` ($k = 2, \kappa = 1$) with the existing `tanh2` model (mapping $M = \sqrt{6\alpha} M_P$, $\Lambda^4 = \frac{3}{2}\lambda M_P^4$); resolution studies; long runs.

8 Redshifting the GW spectrum to today

The lattice gives the GW spectrum at the end of the run, t_f , as a function of the comoving wavenumber in program units:

$$\Omega_{\text{GW}}^{(f)}(k) \equiv \frac{1}{\rho_f} \left(\frac{d\rho_{\text{GW}}}{d \ln k} \right)_f, \quad (13)$$

where ρ_f is the total energy density at t_f (file `spectra_energy_gws.txt`, columns $\tilde{k}$, Ω_{GW} , multiplicity). To compare it with detectors we need the frequency today, f , of each mode k , and the spectrum today, $h^2\Omega_{\text{GW}}(f)$. This is done in Sec. 5 of `runs/attractor_runs.ipynb` (code in `runs/make_notebook.py`, function `gw_today`). We follow Figueroa & Torrenti, arXiv:1707.04533, Sec. 2.1, Eqs. (2.16)–(2.18); the derivation is repeated here so that the conventions and the treatment of a time-dependent w are explicit.

8.1 Times, notation and assumptions

- t_i : start of the simulation (end of inflation). COSMOLATTICE sets $a_i = 1$, so the comoving k equals the physical wavenumber at t_i .
- t_f : end of GW production, taken as the time of the spectrum being redshifted (the last output, $\tilde{t}_f = 290$, unless stated otherwise).

- t_{RD} : onset of radiation domination (RD). From then on the universe is a thermal plasma with entropy conserved.
- t_o : today. g_t and $g_{s,t}$ are the relativistic and entropic degrees of freedom at time t .
- Between t_i and t_{RD} the equation of state $w = p/\rho$ is arbitrary (and time dependent); $\rho \propto a^{-3(1+w)}$ for constant w , and in general

$$\frac{d \ln \rho}{d \ln a} = -3(1+w). \quad (14)$$

- After production, GWs are free waves well inside the horizon: their comoving k is fixed, and their energy redshifts as radiation, $d\rho_{\text{GW}}/d \ln k \propto a^{-4}$ for each k .

8.2 Frequency today

A comoving mode k has physical wavenumber k/a . Its frequency today is

$$f = \frac{1}{2\pi} \frac{k}{a_o} = \frac{a_i}{a_o} \frac{k}{2\pi}, \quad \frac{a_i}{a_o} = \frac{a_i}{a_{\text{RD}}} \frac{a_{\text{RD}}}{a_o}. \quad (15)$$

From t_{RD} to today. Entropy conservation, $g_s T^3 a^3 = \text{const}$, gives $a_{\text{RD}}/a_o = (g_{s,o}/g_{s,\text{RD}})^{1/3} T_o/T_{\text{RD}}$. With $\rho = \frac{\pi^2}{30} g T^4$ for radiation, $T_o/T_{\text{RD}} = (g_{\text{RD}}/g_o)^{1/4} (\rho_o/\rho_{\text{RD}})^{1/4}$, where ρ_o is the radiation energy density today. Hence

$$\frac{a_{\text{RD}}}{a_o} = \left(\frac{g_{s,o}}{g_{s,\text{RD}}} \right)^{1/3} \left(\frac{g_o}{g_{\text{RD}}} \right)^{-1/4} \left(\frac{\rho_o}{\rho_{\text{RD}}} \right)^{1/4}. \quad (16)$$

From t_i to t_{RD} . For constant w , $\rho_{\text{RD}} = \rho_i (a_i/a_{\text{RD}})^{3(1+w)}$, so $\rho_{\text{RD}}^{-1/4} = \rho_i^{-1/4} (a_i/a_{\text{RD}})^{-3(1+w)/4}$. Inserting this and Eq. (16) in Eq. (15), the powers of a_i/a_{RD} combine as $1 - \frac{3}{4}(1+w) = \frac{1}{4}(1-3w)$:

$$\frac{a_i}{a_o} = \epsilon_i^{1/4} \left(\frac{g_{s,o}}{g_{s,\text{RD}}} \right)^{1/3} \left(\frac{g_o}{g_{\text{RD}}} \right)^{-1/4} \left(\frac{\rho_o}{\rho_i} \right)^{1/4}, \quad \epsilon_i \equiv \left(\frac{a_i}{a_{\text{RD}}} \right)^{1-3w}. \quad (17)$$

This is the paper's Eq. (2.16). $\epsilon_i < 1$ for $w < 1/3$ (shift to the IR), $= 1$ for $w = 1/3$, > 1 for a stiff $w > 1/3$.

Time-dependent w . Integrating Eq. (14), $\rho_{\text{RD}}/\rho_i = \exp[-3 \int_{a_i}^{a_{\text{RD}}} (1+w) d \ln a]$. Then

$$\frac{a_i}{a_{\text{RD}}} \left(\frac{\rho_i}{\rho_{\text{RD}}} \right)^{-1/4} = \exp \left[- \int d \ln a + \frac{3}{4} \int (1+w) d \ln a \right] = \exp \left[- \frac{1}{4} \int_{a_i}^{a_{\text{RD}}} (1-3w) d \ln a \right], \quad (18)$$

so Eq. (17) holds with

$$\epsilon_i = \exp \left[- \int_{a_i}^{a_{\text{RD}}} (1-3w) d \ln a \right]. \quad (19)$$

For constant w this reduces to $(a_i/a_{\text{RD}})^{1-3w}$.

Result. The frequency today is (paper's Eq. (2.17), first line)

$$f = \epsilon_i^{1/4} \left(\frac{g_{s,o}}{g_{s,\text{RD}}} \right)^{1/3} \left(\frac{g_o}{g_{\text{RD}}} \right)^{-1/4} \left(\frac{\rho_o}{\rho_i} \right)^{1/4} \frac{k}{2\pi} \quad (20)$$

With the constants of Sec. 8.4 the numerical prefactor is

$$f \simeq 4.6 \times 10^{10} \epsilon_i^{1/4} \frac{k}{\rho_i^{1/4}} \text{ Hz}, \quad (21)$$

which reproduces the standard $f \simeq 4 \times 10^{10} (k/\rho_i^{1/4}) \text{ Hz}$ of Dufaux et al. (2007) for $\epsilon_i = 1$.

A 2π in the paper. The second line of the paper's Eq. (2.17), $f \simeq \epsilon_i^{1/4} (k/\rho_i^{1/4}) \times 8 \cdot 10^9 \text{ Hz}$, does not follow from its first line. With the paper's $\rho_o \simeq 2 \cdot 10^{-15} \text{ eV}^4$ and the g factors set to 1, the first line gives $\rho_o^{1/4}/(2\pi) = 5.1 \times 10^{10} \text{ Hz}$, which is 2π times $8 \cdot 10^9 \text{ Hz}$. We use the first line, Eq. (20).

8.3 Amplitude today

After t_f , $d\rho_{\text{GW}}/d\ln k \propto a^{-4}$ at fixed k , so

$$\Omega_{\text{GW}}(k, t_o) = \frac{1}{\rho_c} \left(\frac{d\rho_{\text{GW}}}{d\ln k} \right)_o = \frac{1}{\rho_c} \left(\frac{a_f}{a_o} \right)^4 \rho_f \Omega_{\text{GW}}^{(f)}(k), \quad (22)$$

with ρ_c the critical density today. Write $\rho_f a_f^4$ in terms of the radiation density today, $\rho_o = \rho_c \Omega_{\text{rad}}$:

$$\frac{\rho_f a_f^4}{\rho_o a_o^4} = \underbrace{\frac{\rho_f a_f^4}{\rho_{\text{RD}} a_{\text{RD}}^4}}_{(a_f/a_{\text{RD}})^{1-3w}} \underbrace{\frac{\rho_{\text{RD}} a_{\text{RD}}^4}{\rho_o a_o^4}}_{(g_{\text{RD}}/g_o)(g_{s,o}/g_{s,\text{RD}})^{4/3}}. \quad (23)$$

The first factor follows from $\rho_{\text{RD}} = \rho_f (a_f/a_{\text{RD}})^{3(1+w)}$, so that a^4 against $a^{3(1+w)}$ leaves the power $1 - 3w$; for time-dependent w it is

$$\epsilon_f \equiv \left(\frac{a_f}{a_{\text{RD}}} \right)^{1-3w} \rightarrow \exp \left[- \int_{a_f}^{a_{\text{RD}}} (1 - 3w) d\ln a \right]. \quad (24)$$

The second factor uses $\rho a^4 \propto g T^4 a^4 \propto g g_s^{-4/3}$ during the thermal phase. Multiplying by h^2 (paper's Eq. (2.18), first line):

$$\boxed{h^2 \Omega_{\text{GW}}(f) = h^2 \Omega_{\text{rad}} \epsilon_f \left(\frac{g_{s,o}}{g_{s,\text{RD}}} \right)^{4/3} \left(\frac{g_{\text{RD}}}{g_o} \right) \Omega_{\text{GW}}^{(f)}(k)} \simeq 1.6 \times 10^{-5} \epsilon_f \Omega_{\text{GW}}^{(f)}(k). \quad (25)$$

$\epsilon_f < 1$ (suppression) if there is a phase with $w < 1/3$ between the end of production and RD. Since Ω_{GW} is defined per logarithmic interval and $f \propto k$ at fixed time, $d\ln f = d\ln k$: the spectrum is mapped point by point, $h^2 \Omega_{\text{GW}}(f(k))$, with no Jacobian.

Note the difference between the two factors: ϵ_i (frequency) involves the whole history from t_i , because k is defined at t_i ; ϵ_f (amplitude) only the history after t_f , because $\Omega_{\text{GW}}^{(f)}$ is normalized at t_f . They are related by $\epsilon_f = \epsilon_i \exp[\int_{a_i}^{a_f} (1 - 3w) d\ln a]$.

8.4 Implementation

Lattice quantities in physical units. With $f_* = M_P$ and $a_i = 1$:

$$k [\text{GeV}] = \tilde{k} \omega_*, \quad \rho_i = \tilde{E}_{\text{tot}}(\tilde{t} = 0) M_P^2 \omega_*^2, \quad \omega_* = \sqrt{\lambda_k} M_P x_i^{(k-2)/2}, \quad (26)$$

where $\tilde{E}_{\text{tot}}$ is read from `average_energies.txt` and ω_* is recomputed from `run.in` exactly as in the model headers (Sec. 4). Conversion to Hz: in natural units Eq. (20) gives f in GeV, and $f [\text{Hz}] = f [\text{GeV}]/\hbar$ with $1 \text{ GeV}/\hbar = 1.519267 \times 10^{24} \text{ s}^{-1}$.

The expansion factors. The integral $\int (1 - 3w) d\ln a$ is split at t_f :

$$\epsilon_i = \exp \left[- (I_f + I_{\text{post}}) \right], \quad \epsilon_f = \exp \left[- I_{\text{post}} \right], \quad (27)$$

- $I_f = \int_{a_i}^{a_f} (1 - 3w) d\ln a$ is computed *from the lattice*. The equation of state is $w = p/\rho$ with the volume-averaged energies of the two scalars, $\rho = K + G + V$ and $p = K - G/3 - V$ (K kinetic, G gradient, V potential; GWs are negligible). $\ln a$ is interpolated from `average_scale_factor.txt` to the times of `average_energies.txt`, and the integral is done with the trapezoidal rule. Integrating the instantaneous w automatically averages over the inflaton oscillations.

- $I_{\text{post}} = (1 - 3w_{\text{post}})N_{\text{post}}$ describes the unknown history between the end of the run and the onset of RD: $N_{\text{post}} = \ln(a_{\text{RD}}/a_f)$ e -folds at constant w_{post} (notebook variables `N_POST`, `W_POST`). The default $N_{\text{post}} = 0$ (instant RD at t_f) gives $\epsilon_f = 1$. Since $w \simeq 0.24 < 1/3$ at the end of the runs, this is an upper bound on both f and $h^2\Omega_{\text{GW}}$.

Constants. $T_0 = 2.7255 \text{ K}$, $g_o = 3.36$, $g_{s,o} = 3.91$, so $\rho_o = \frac{\pi^2}{30}g_o T_0^4 = 3.36 \times 10^{-15} \text{ eV}^4$; $h^2\Omega_{\text{rad}} = 4.15 \times 10^{-5}$, which is consistent with this ρ_o ($\rho_c/h^2 = 8.10 \times 10^{-47} \text{ GeV}^4$); $g_{\text{RD}} = g_{s,\text{RD}} = 106.75$ (full SM). The g factors are $(g_{s,o}/g_{s,\text{RD}})^{1/3}(g_o/g_{\text{RD}})^{-1/4} = 0.79$ in Eq. (20) and $(g_{s,o}/g_{s,\text{RD}})^{4/3}(g_{\text{RD}}/g_o) = 0.39$ in Eq. (25). $M_P = 2.435 \times 10^{18} \text{ GeV}$.

Spectra at intermediate times. `gw_today(name, idx)` can redshift the spectrum of any output time, treating that time as t_f . Then I_f , and so the $k \rightarrow f$ map, changes with t_f . The same map $f/\tilde{k}$ is used to plot the field spectra against the frequency today.

8.5 Results for the two runs

Runs `runs/E_starobinsky_gw` and `runs/T_k2_a1_gw` ($N = 64$, $k_{\text{IR}} = 0.5$, $q_0 \approx 10^4$), redshifted from $\tilde{t}_f = 290$ with $N_{\text{post}} = 0$:

Run	ω_* [GeV]	$\rho_i^{1/4}$ [GeV]	a_f	ϵ_i	$f/\tilde{k}$ [Hz]	f_p [Hz]	$h^2\Omega_{\text{GW}}(f_p)$
E, Starobinsky, $g = 4.5 \times 10^{-3}$	3.35×10^{13}	5.84×10^{15}	18.9	0.126	1.57×10^8	2.6×10^9	1.6×10^{-9}
T, $k = 2$, $g = 1.7 \times 10^{-3}$	1.76×10^{13}	5.47×10^{15}	26.1	0.094	8.16×10^7	1.35×10^9	2.1×10^{-9}

$\epsilon_i \approx 0.1$ corresponds to an average $\langle w \rangle \approx 0.1$ over the ~ 3 e -folds of the run, i.e. a frequency shift by $\epsilon_i^{1/4} \approx 0.6$ relative to instant RD at t_i . A matter phase of $N_{\text{post}} = 2$ e -folds after the run lowers f by $e^{-1/2} \approx 0.61$ and $h^2\Omega_{\text{GW}}$ by $e^{-2} \approx 0.14$ (dashed curves in the notebook). These numbers are not converged: the GW peak ($\tilde{k} \simeq 17$) is close to the lattice cutoff $\tilde{k}_{\text{max}} \simeq 28$, so the runs must be repeated at $N = 128$ – 256 before quoting them.

9 How to build and run

```
export PATH=~/.miniconda3/envs/cosmolattice/bin:$PATH # provides cmake and FFTW
cd ~/cosmolattice
mkdir build_attractorE && cd build_attractorE
cmake .. -DMODEL=attractorE && make -j8
./attractorE input=./models/parameter-files/attractorE_k2_a1_starobinsky.in
# any key can be overridden on the command line, e.g. N=128 tMax=500 g=1e-4
```

For the T-model replace `attractorE` with `attractorT`. To make a new benchmark, run the helper with the desired (α, k, κ, N_*) and copy its output into a copy of one of the existing `.in` files.

10 Open points

- Fix N_* self-consistently: for $k \geq 6$ measure the fragmentation time and temperature on the lattice and feed them back into the paper’s App. B expression for N_* , which depends on $\max[T_{\text{RH}}, T_{\text{frag}}]$.
- Choose the daughter coupling g (and possibly trilinear or fermionic decay channels) for the reheating study.
- For $k \geq 6$, after fragmentation the equation of state tends to $w = 1/3$; consider $\alpha_t = 1$ for long runs.

- Tune lattice settings (Sec. 6) with resolution tests.